

AYE
TECH HUB

• MEP ENGINEERING • LESSON 05

Heating Systems and Boilers

Combi · System · Condensing · Hydronic · Underfloor

Lesson 05 of 30 | MEP Engineering Program

Awet G. Nway

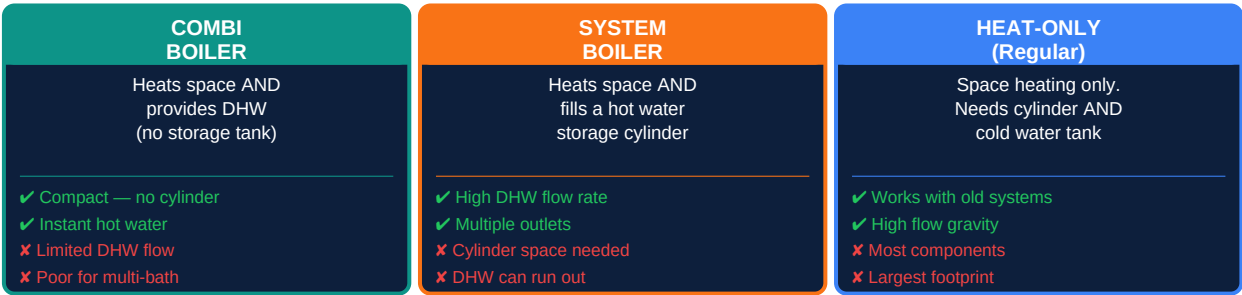
Founder, AYE Tech Hub | MEP & Building Services Engineering Specialist

- Boiler types: combi, system, heat-only, condensing — and how to choose
- Hydronic heating circuit: boiler, pump, expansion vessel, radiators, TRVs
- Two-pipe vs one-pipe heating systems — layout, pros, cons, and application
- Flow and return temperatures, ΔT , heat emitter sizing fundamentals
- Controls: thermostats, zone valves, thermostatic radiator valves (TRVs)

Heating Systems — Keeping Buildings Warm

In MEP engineering, the heating system is the M (Mechanical) side of the building. Its job is to maintain comfortable temperatures for occupants and protect the building from freezing. The heart of most heating systems is a **boiler** — a device that burns fuel (or uses electricity) to heat water, which is then pumped around the building to heat emitters (radiators, underfloor pipes, fan coil units). This is called a **hydronic** (wet) heating system.

Figure 1 — Three Boiler Types: Combi / System / Heat-Only — Selection Guide



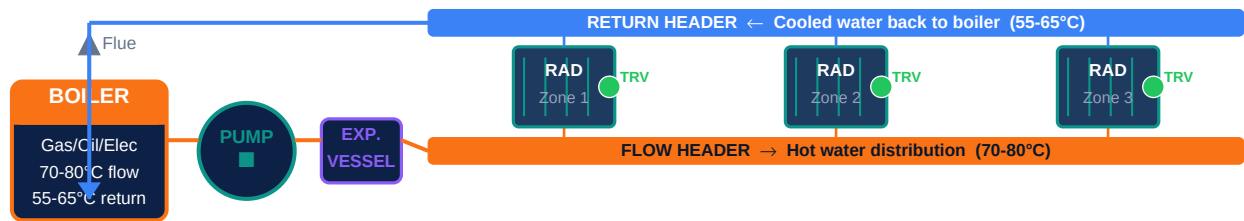
Boiler Type	How It Works	Key Components	Best Application
Combi (Combination)	Single unit heats both space heating water and domestic hot water (DHW) on demand — no storage cylinder needed. DHW takes priority when a tap opens.	Burner, heat exchanger, diverter valve, built-in pump, expansion vessel, PRV	Flats, small homes (1-2 bathrooms). Where space is limited. Gas combi most common in East Africa commercial projects.
System Boiler	Heats space AND heats water stored in a hot water cylinder. All major components built in — only external cylinder needed.	Burner, heat exchanger, pump, expansion vessel, PRV, unvented cylinder	Medium homes (2-4 bathrooms). Where multiple simultaneous hot water outlets are needed.
Heat-Only (Regular / Conventional)	Heats space only — works with separate cold water storage tank and hot cylinder. Gravity fed, open-vented system.	Burner, heat exchanger, separate pump, F&E; tank, cylinder, open vent pipe	Older buildings with existing systems. Large properties where gravity systems work well.
Condensing Boiler	A heat exchanger captures heat from flue gases (which would otherwise be wasted). The water vapour in exhaust condenses, releasing extra latent heat.	Secondary heat exchanger, condensate trap and drain, plastic flue (lower temp)	Required by law in UK and most of Europe for new installations. Efficiency up to 95% (vs 75-80% for non-condensing).

Condensing boiler efficiency explained: A non-condensing boiler exhausts hot flue gases at 180-200°C — wasting 15-25% of the fuel energy up the flue. A condensing boiler cools the flue gases to 50-55°C, recovering that heat into the return water. This only works efficiently when the return water temperature is below 55°C (so specify low-temperature systems — 60/40°C instead of 80/60°C to maximise condensing mode operation).

The Hydronic Heating Circuit

A hydronic (water-based) heating system circulates hot water from the boiler through pipes to heat emitters (radiators, underfloor heating, fan coil units) and back to the boiler to be reheated. Understanding each component in the circuit is essential for designing, commissioning, and troubleshooting MEP heating systems.

Figure 2 — Hydronic Heating Circuit: Boiler → Pump → Expansion Vessel → Radiators (with TRVs) → Return

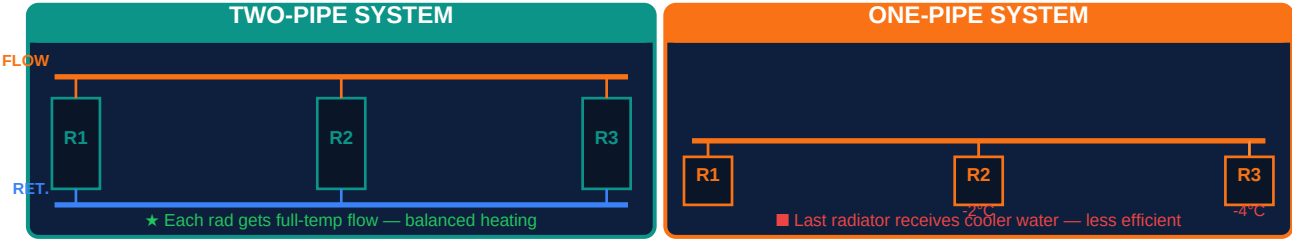


Component	Function	Key Design Parameter
Boiler	Heats water by burning gas/oil or using electric elements. Output measured in kW. Must match peak heat loss of building.	Output kW ≥ building heat loss (kW). Typical: 24-35 kW for houses, 50-500 kW for commercial.
Circulating Pump	Forces hot water around the system at the required flow rate. Without it, water would not circulate (gravity systems are passive but slow).	Size for system flow rate Q (L/s) and available head (kPa). Use variable-speed pumps for efficiency.
Expansion Vessel	Water expands when heated — this vessel absorbs the volume increase and maintains system pressure within safe limits.	Size for system water volume × expansion coefficient. Pre-charge pressure = static head of system.
Pressure Relief Valve (PRV)	Safety device — opens automatically if system pressure exceeds the maximum safe pressure, discharging water to drain.	Set to 3 bar for most residential systems. Must discharge to safe location.
Thermostatic Radiator Valve (TRV)	Senses room air temperature and modulates radiator flow accordingly. Provides individual room temperature control without a zone valve.	Fit to flow side of radiator. Do not fit to boiler room radiator — leave fully open as reference.
Zone Valve	Motorised valve that opens/closes a heating zone on demand from room thermostat or programmer. Controls separate zones (upstairs/downstairs).	2-port for zone isolation. 3-port for diverting between heating and hot water (combi alternative).
Flow/Return Temperatures	Flow (from boiler) is hotter. Return (back to boiler) is cooler. ΔT = flow – return temperature (typically 10-20°C).	Standard: 80°C flow / 60°C return (ΔT=20°C). Low-temp: 60°C / 40°C for condensing efficiency.

Pipe Systems, Heat Emitters, and Controls

The layout of the pipe distribution network determines how evenly heat reaches every room. The two fundamental approaches — two-pipe and one-pipe — have very different performance characteristics.

Figure 3 — Two-Pipe vs One-Pipe Heating Systems: Layout, Temperature, and Efficiency



Heat Emitter Type	How It Works	Flow Temp Needed	Best For
Panel Radiator	Hot water flows through steel panels. Heat transferred to room by radiation (~30%) and convection (~70%). Most common emitter in buildings worldwide.	70-80°C (standard) 55-65°C (low-temp)	Residential and commercial buildings. Simple, reliable, low maintenance. Wall-mounted, visible.
Underfloor Heating (UFH)	Plastic pipes embedded in screed or between floor joists carry warm water. Heat radiates upward from entire floor surface.	35-45°C — very low (ideal for condensing)	New-build homes, open-plan offices, luxury fit-outs. Invisible, comfortable, but slow response.
Fan Coil Unit (FCU)	A small coil of hot water pipe with a fan blowing air through it. Provides both heating and cooling (4-pipe version).	60-70°C heating mode 7-12°C chilled mode	Hotel rooms, apartments, office perimeters. Fast response, individual room control. Requires maintenance (filters).
Convactor Heater	Finned-tube heat exchanger — larger surface area than radiator for higher output. Natural convection (no fan).	70-80°C standard	High-ceiling spaces where radiators would be impractical. Corridors, stairwells.

Control Component	Function	Placement	MEP Design Note
Room Thermostat	Senses air temperature and calls for heat when below setpoint. Turns zone valve on.	Representative room — not in draughts, direct sunlight, or next to heat sources.	1 per zone minimum. Smart thermostats (Nest, Hive) allow remote and schedule control.
Boiler Controller / Programmer	Sets heating and hot water on/off schedules. Communicates with boiler and zone valves.	Boiler or central accessible location.	OpenTherm protocol allows modulating boilers to match demand — more efficient than on/off.
Thermostatic Radiator Valve	Individual room temperature control. Closes when room is warm enough.	Flow connection on each radiator.	Prevents wasted energy in unused rooms. Do NOT fit to the room with the main thermostat.
Weather Compensation Control	Adjusts boiler flow temperature based on outdoor temperature. Cold outside → high flow temp. Mild → lower flow temp.	Outdoor temperature sensor on north-facing wall.	Increases condensing boiler efficiency by 5-10%. Required for A-rated energy performance in many codes.

What comes next — MEP-006: Cooling Systems and Chillers — how chilled water systems work, chiller types (air-cooled vs water-cooled), cooling towers, chilled water distribution (primary/secondary circuits), fan coil units in cooling mode, and chiller plant sizing and COP.